

Gradient ascent for logistic regression with L2 penalty

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1 Likelyhood function

$$L(w) = \prod_{i=0}^{n-1} P(y_i|x_i, w),$$

2 Log likelyhood function

$$\sum_{i=0}^{n-1} ((\mathbb{1}[y_i = +1] - 1)w^\top h(x_i) - \ln(1 + e^{-w^\top h(x_i)}))$$

From this we subtract λ times L2 norm of coefficients

$$\|w\|_2^2 = \sum_{j=1}^{d-1} w_j^2$$

(0-th coefficient (j=0) is not included here) Which will give the final formula

$$\ell(w) = \sum_{i=0}^{n-1} ((\mathbb{1}[y_i = +1] - 1)w^\top h(x_i) - \ln(1 + e^{-w^\top h(x_i)})) - \lambda \|w\|_2^2$$

3 Derivative of log of the likelihood function

$$\sum_{i=0}^{n-1} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w))$$

From this we subtract

$$2\lambda w_j$$

to account for the L2 penalty which will give the final formula

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i=0}^{n-1} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w)) - 2\lambda w_j$$

if $j > 0$

$$\frac{\partial \ell(w)}{\partial w_j} = \sum_{i=0}^{n-1} h_j(x_i)(\mathbb{1}[y_i = +1] - P(y_i = +1|x_i, w))$$

if $j = 0$